Paleomagnetism of the Contact Aureole and Late Dikes of the Otto Stock, Ontario, and its Application to Early Proterozoic Apparent Polar Wandering[1]

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The Otto Stock consists mainly of syenite. Except for some mafic phases the syenites are magnetically unstable, but late-stage lamprophyre dikes and basic rocks in the contact aureole contain abundant stable material. Their mean direction of magnetization is 330°, + 71° (18 sites, 84 samples, $\alpha_{95} = 3°$) with a corresponding pole at 69°N, 133°W ($A_{95} = 5°$). A subordinate later magnetization also occurs in three lamprophyre dikes. Using these and previous results a path of apparent polar wander for Laurentia for the interval 2200 to 1800 m.y. of simple form is constructed. The fact that a single simple polar path can be constructed, which is consistent with data from localities spread widely over the Laurentian Shield, indicates that the Hudsonian orogeny may not have been accompanied by very large relative motions of the bordering Archean cratons.

Le stock d'Otto consiste principalement de syenite. Excepté pour certaines phases mafiques les syénites sont magnétiquement instables, mais les phases tardives de filons de lamprophyre et de roches basiques dans l'auréole de contact contiennent abondamment du matériel stable.

Leur direction moyenne de magnétisation est de 330°, + 71° (18 sites, 84 échantillons, $\alpha_{95} = 3°$) avec un pôle correspondant à 69°N, 133°W ($A_{95} = 5°$). Une magnétisation inférieure s'est produite aussi dans trois filons de lamprophyre.

Avec ces résultats et ceux obtenus précédemment un tracé de forme simple montrant un pôle apparent errant entre 2200 et 1800 m.a. a été construit pour Laurentia.

Le fait qu'un seul tracé ait été construit, qui est d'ailleurs constant avec les résultats provenant des localités distribuées largement sur le bouclier Laurentien, indique que l'orogène Hudsonienne pourrait ne pas avoir été accompagné de mouvements de cratons Archéens relativement larges en bordure du Bouclier. [Traduit par le journal]

Geology and Sampling

The Otto stock is situated in the Superior Structural Province near Kirkland Lake, Ontario (Fig. 1). Figure 2 is a schematic representation of its relationship with adjacent rock units and Fig. 3 is a map of the body showing sample locations. The stock is about 10 km in diameter and was intruded into Archean rocks. Its geology has been described by Lovell (1972). The stock has yielded an Rb–Sr whole-rock isochron of 2160 ± 80 m.y. (K. Bell, personal communication, 1975).[2] Earlier radiometric work is reviewed in that paper. The Archean country rock consists mainly of metavolcanics and felsic intrusions, both of which were intruded by unmetamorphosed Matachewan diabase dikes, whose age is about 2600 m.y. (Gates and Hurley 1973). The latter are not observed to cut the Otto stock, although a dike that probably belongs to the Matachewan swarm, occurs in the contact aureole (site 10 on Fig. 3) that extends around the stock. The aureole is about 300 m wide, and within it the metamorphic grade reaches amphibolite facies. The stock itself is unmetamorphosed, and consists of syenite, locally contaminated by country rock. The effects of contamination range from slightly digested basic zenoliths, through diorite, to mafic syenite and lamprophyre (Lovell 1972). Lamprophyre dikes, generally a few metres in width, are abundant along Highway 112. In the center-west of the stock along Highway 11 is a large outcrop of basic rock, which is considered by Lovell to be a roof pendant or inclusion. This outcrop has been heated and metamorphosed to amphibolite facies by the intrusion of the Otto stock. It is referred to here as the *west-central baked zone*. The contact aureole, the west-central baked zone, and the lamprophyre dikes are obvious targets for paleomagnetic studies, because they might be expected to record the field at the time of intrusion.

To the south and west of the Otto stock is the

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[2] K. Bell and J. Blenkinsop. "A Rb–Sr whole-rock isochron from the Otto stock, Ontario." Manuscript submitted to the Can. J. Earth Sci., June 1975.

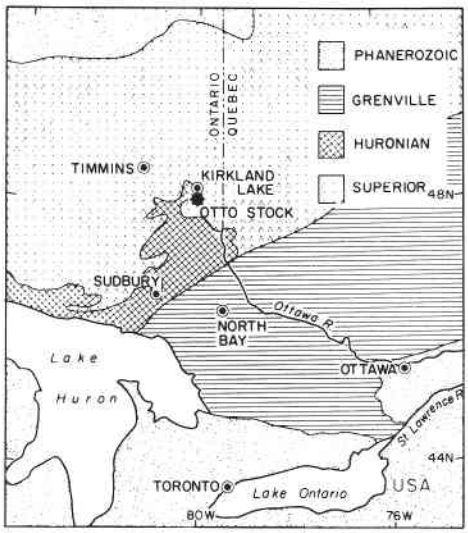


FIG. 1. Location of the Otto stock in the Superior Structural Province.

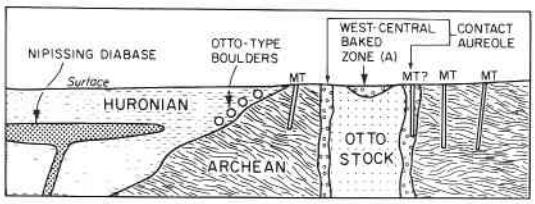


FIG. 2. Schematic relationship of the Otto stock to other rock units in the vicinity. MT are Matachewan dikes. The stock and its contact aureole are grossly exaggerated.

main outcrop of Huronian sedimentary rocks intruded by Nipissing diabase. Whole-rock Rb–Sr ages of 2150 ± 50 m.y. (Van Schmus 1965) and 2160 ± 60 m.y. (Fairbairn *et al.* 1969) have been obtained from Nipissing diabase. The Otto stock and the Huronian are not in contact, but conglomerates in the Huronian (Gowganda conglomerate of the Cobalt Group) contain boulders comparable to certain phases of the stock (Lovell 1972). This indicates (but does not prove) that the stock is contemporaneous with, or older than the upper part of the Huronian. The probable sequence of rock units is given below from youngest to oldest based mainly on Lovell (1972).

1. Nipissing diabase (intrudes Huronian).

2a. Huronian (overlies Archean and contains Otto-type boulders).

2b. Otto stock (intrudes Archean, but is not seen to be cut by Matachewan dikes; relationship to Huronian uncertain, see above).

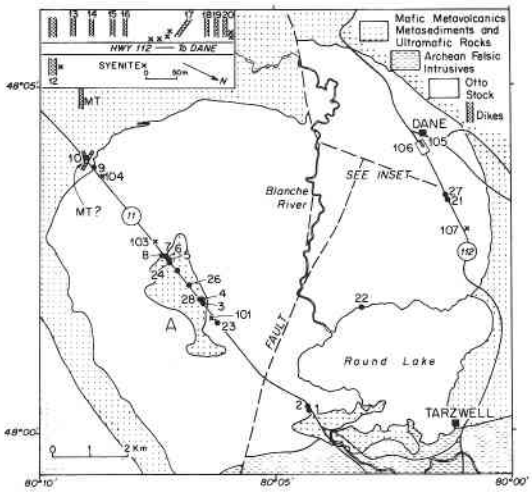


FIG. 3. Locality sketch-map of the Otto stock. The paleomagnetic sampling sites are marked by dots and numbered 1 to 28. The sampling sites for radiometric work are marked by crosses and numbered 101 to 107 (Bell and Blenkinsop, see footnote 2). The lamprophyre dikes near Dane are shown in the inset. The crosses in the inset are syenite samples referred to collectively in the text as site 11.

3. Matachewan dikes (cut Archean).

4. Archean.

The Rb–Sr ages of the Otto stock and the Nipissing diabase are statistically indistinguishable (Bell and Blenkinsop)[3] indicating that events '2b' and '1' span approximately 100 m.y. or less.

The paleomagnetic sampling sites are marked on Fig. 3. Much of the area is wooded, access is difficult, and natural exposures are commonly weathered. Sampling was confined to highway cuttings and the lake shore where unweathered rock is available. Cores were obtained from 28 sites. Two cylindrical specimens were cut from each core. The magnetization was measured by a Schonstedt SM1 magnetometer. Alternating field (AF) and thermal demagnetization was carried out using the apparatus described by Roy *et al.* (1972, 1973). The initial magnetization (NRM) directions in all specimens are plotted in Fig. 4. The most notable features are a strong condensation of directions not far from the present field, superimposed on a wide scatter of directions.

Magnetism of Syenite

Six sites in syenite were sampled. Magnetization intensities (NRM) are about 100 microgauss

[3]See footnote 2.

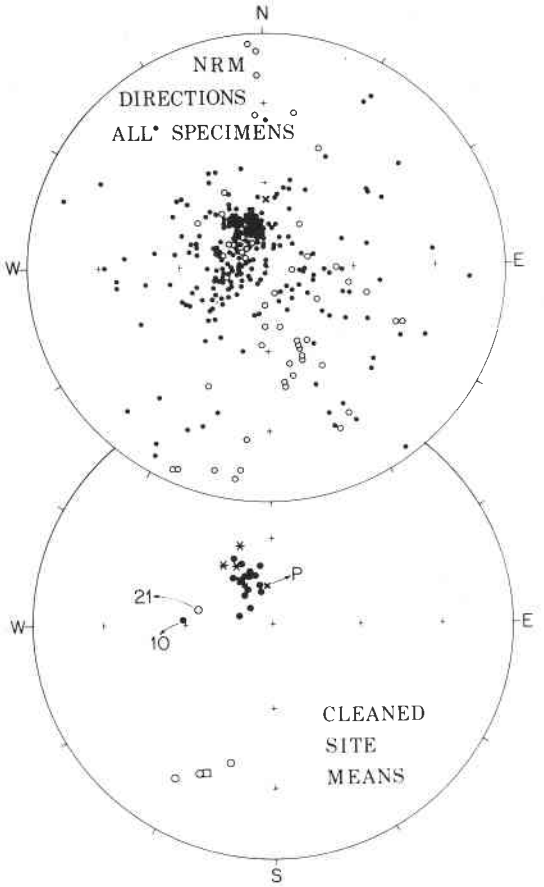


FIG. 4. Above the directions of natural remanent magnetization (NRM) in all specimens are plotted. Dots are downward and circles are upward inclinations. Below, the directions at those sites with stable magnetization are given. The stars are the *R* magnetizations reversed into the group of *N* magnetizations (dots). The open symbols (upward inclination) are the *S* magnetization. *P* is the present field direction. The perimeter is the present horizontal. The squares are the means of the *S* magnetizations, and of the *N* and *R* magnetizations averaged without respect to sign.

(μG). Sites 1, 2, and 22 are in pink syenite containing dark mafic material, sites 8 and 11 in homogenous pink syenite, site 3 in a brownish syenite dike at the margin of the west-central baked zone, and site 23 is in mafic syenite. Except for sites 11 and 23, the initial directions were scattered about the present field, the intensities fall sharply during AF and thermal demagnetization (Fig. 5) and the directions became random. Polished sections have only large irregular grains up to 1000 μm in diameter, and smaller euhedral grains typically 50–20 μm in size of fresh magnetite with no visible intergrowths. Such grains are probably multidomain and unstable magnetically, and account for the unsatisfactory results from these syenites. At site 11 the directions during demagnetization remain scattered, but move significantly away from the present field with direction toward the southeast with downward inclination (Table 1). This magnetization is only moderately stable and the direction is characteristic of post-Hudsonian overprints found widely in lower Proterozoic rocks (McGlynn *et al.* 1974; McGlynn and Irving 1975). The gray mafic syenite at site 23 has a stable magnetization which is described later. Because of the generally unstable magnetization of the syenites later sampling was concentrated in basic rocks from the west-central baked zone, the contact aureole, and the lamprophyre dikes.

West-central Baked Zone

Samples from basic rocks in the west-central baked zone were obtained at 8 sites along Highway 11 (Fig. 3). The intensities of NRM are typically about 100 μG. Site 7 is a lamprophyre dike that is magnetically unstable. The

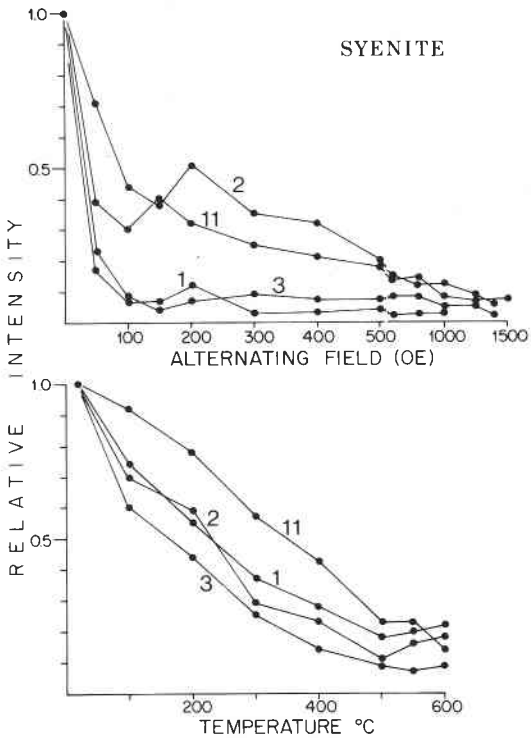


FIG. 5. AF (above) and thermal (below) demagnetization curves for syenite from sites 1, 2, 3, and 11.

TABLE 1. Site mean directions

Site	*A*	*C*	*n*	*M*	*D*,*I*	*R*	*k*	Procedure
4	W	N	5	71	336, +70	4.99	404	500 Oe, 500 °C
5	W	N	5	143	334, +71	4.99	434	300–400 Oe
6	W	N	6	114	347, +69	5.97	156	200–300 Oe, 500 °C
10	MT	N	11	859	273, +59	9.71	8	550 °C
11	S	—	8	161	124, +65	6.62	5	300 Oe, 500 °C
12	L	N	5	5	320, +69	4.94	66	500 °C
13	L	N	5	128	323, +73	4.99	440	400–500 Oe
14	L	S	4	33	205, −30	3.92	39	400 Oe, 500 °C
15	L	N	5	17	331, +71	4.73	15	400 Oe
16	L	N	5	18	317, +77	4.99	1074	300 Oe
17	L	N	5	26	343, +76	4.69	13	550 °C
18	L	N	5	15	304, +80	4.97	120	500 °C
19*N*	L	N	6	24	341, +78	5.97	176	550 °C
19*S*	L	S	7	—	197, −38	6.86	44	*V*, 300–400 °C
20*N*	L	N	5	8	293, +78	4.83	24	550 °C
20*S*	L	S	5	—	212, −23	4.88	34	*V*, 400–500 Oe, 300–400 °C
21	L	?	5	361	291, −64	4.92	48	100–500 Oe
23	MS	R	4	28	158, −60	3.98	197	200–600 Oe
24	W	N	5	165	339, +72	4.99	2142	200 Oe
25	W	N	4	258	322, +75	3.99	285	200 Oe
26	W	R	2	18	151, −66	1.99	—	400 Oe
27	L	N	5	15	332, +66	4.98	214	400 Oe
28*R*	W	R	5	252	141, −63	4.86	28	100 Oe, 500–550 °C
28*N*	W	N	2	—	331, +64	1.99	—	750–1250 Oe

Site numbers are given in the first column. In the second column (*A*) the rocks sampled are indicated: W west-central baked zone, MT Matachewan diabase dike in contact aureole, S syenite, L lamprophyre dike, and MS mafic syenite. In the third column (*C*) the magnetization category is noted (see text). *n* is the number of cores, two specimens measured from each, *M* is the intensity of natural remanent magnetization in microgauss (10^{-6} cgs/cm^3), *D* and *I* are the declination and inclination of the mean direction, *R* is the resultant of *n* core means, and *k* is Fisher's precision (Fisher 1953). In the last column the cleaning conditions are noted, AF in oersted, and temperature of thermal demagnetization in °C. *V* denotes that the directions were obtained by vector analysis (see text).

other six sites show varied behavior, the simplest being the amphibolite at site 5 (Fig. 6). The AF decay curve is smooth, the direction moves away from the present field by about 5°, and then remains unchanged between 400 and 1400 Oe. The thermal decay curve is square shouldered, with blocking temperatures between 550 and 600 °C, and negligible direction variations up to 550 °C. These results show that there is a small unstable magnetization, which is probably a viscous magnetization caused by the present field, and a single very stable magnetization, which is close to the present field but significantly different from it. The stability of the later component is too high to be a viscous component caused by the present field. This very stable magnetization is referred to as the *N* (normal polarity) magnetization.

A further example of such simple behavior, this time from site 4 (specimen 4-1B), is shown in Fig. 7. Again there is a square-shouldered thermal decay curve with blocking temperatures between 550 and 600 °C, negligible directional change, and small but significant departure from the present field. The corresponding AF decay curve is smooth, remanence coercivities up to 1500 Oe are present, and the directional changes are small up to this high field (Fig. 8). The rock at site 4 is a metamorphosed fragmental rock, the basic fragments being a few millimetres in diameter. About 20 m away (site 28) the fragments reach 1 cm in size and the remanence is more complex. Some samples (28-4B on Fig. 7, and 28-3A on Fig. 8) have southerly directions with steep upward inclinations, which are unchanged in direction up to 550 °C and 750 Oe. This magnetization is referred to as the *R* (reversed polarity) magnetization. The thermal decay curves, however, commonly show an increase at 550 °C, which means that a component roughly antiparallel to the *R* magnetization is being removed, indicating the presence of a subordinate *N* magnetization. Other samples show this *N* magnetization in a dramatic way (28-4A on Fig. 8). In alternating fields up to 200 Oe the direction is that of the *R* magnetization, but at increasing fields the direction migrates progressively into the anti-parallel *N* magnetization direction, and remains there between 800 and 1250 Oe. The corresponding

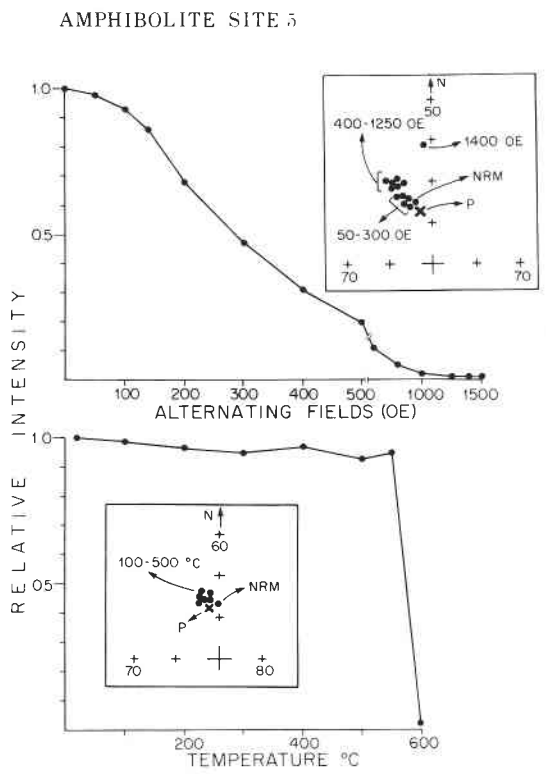


FIG. 6. AF and thermal decay curves for amphibolite at site 5. In the insets the centers of equal-area nets are shown with the directions during demagnetization. Directions all have downward inclination. *P* is the present field.

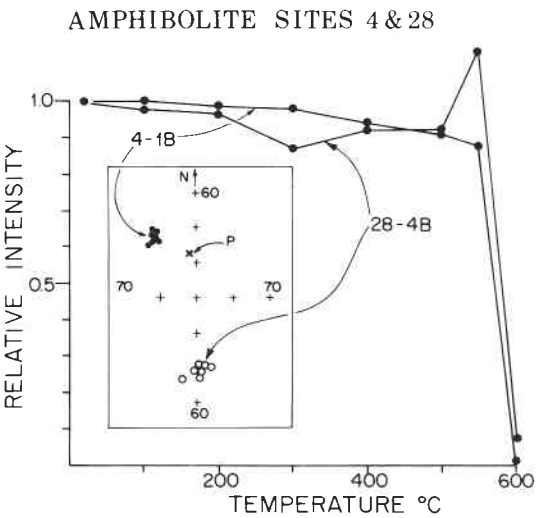


FIG. 7. Thermal decay curves and corresponding direction variations for amphibolite from sites 4 and 28. Dots (circles) denote downward (upward) inclination.

AF curves are shown on Fig. 8. At site 28 both *N* and *R* magnetizations are present in the same specimens, and it is possible to separate the *N* magnetization completely in two samples by AF demagnetization in field between 750 and 1250 Oe. The mean *N* and *R* directions are given in Table 1 and are antiparallel within 4° of one another. The directions at the other 4 sites in the west-central mafic body are listed in Table 1. *N* magnetization occurs at 3 sites (6, 24, and 25), and *R* magnetization at site 26.

In polished section, sulfides, ilmenite, and a few euhedral magnetite grains occur. Very fine-grained (<1 μm) iron minerals, possibly magnetite, occur disseminated, or locally in linear arrays in silicates, notably at site 4. The very stable magnetizations with remanent coercivities in excess of 1000 Oe are presumably owing to

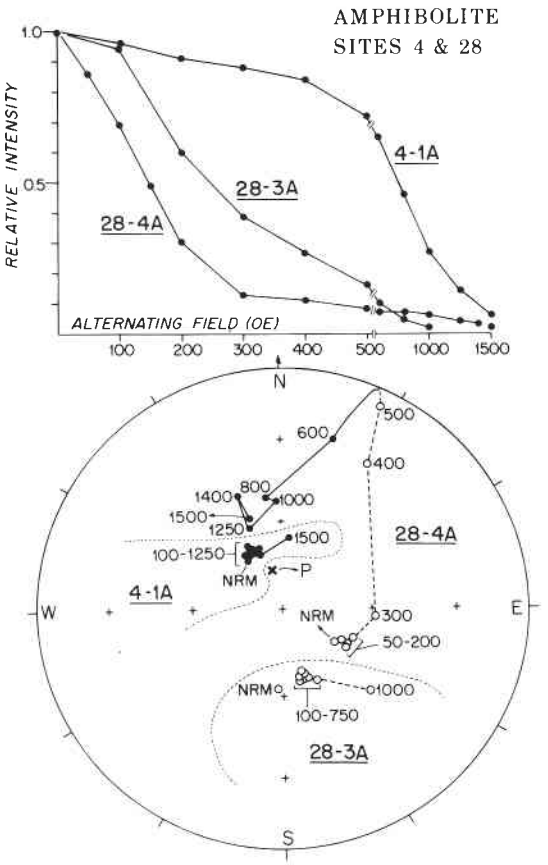


FIG. 8. AF decay curves (above) and corresponding directional changes (below) for amphibolite from sites 4 and 28. Dots (circles) denote downward (upward) inclination.

these grains. The blocking temperatures are predominantly between 550 and 600 °C, suggesting that magnetite was responsible for both the *N* and *R* magnetization. Hematite was not observed. Merrill and Grommé (1969) have described behavior similar to that at site 28 in Cretaceous diorite from the Sierra Nevada, which they attribute to self-reversal. This could be the cause of our results, although it is equally possible that a reversal of the field occurred during slow cooling of the metamorphic aureole.

Lamprophyre Dikes

Twelve lamprophyre dikes intrusive into syenite were sampled. The NRM intensities are low, of the order 10 μG. Site 7 was unstable and had random magnetizations in AF of 100 Oe. The magnetization at sites 12, 13, 15 to 18, and 27 consists of small magnetization along the present field, together with much larger stable *N* magnetizations (Table 1). The blocking temperatures are predominantly between 550 and 650 °C, except at site 17, where 20% of the magnetization is retained at 600 °C. Polished sections have abundant sulfides, and ilmenite, both generally highly altered. There are many euhedral grains of magnetite (10 μm) with no intergrowths, and there are abundant smaller disseminated grains of magnetite. The latter are presumably responsible for the stable *N* magnetization. Hematite occurs at site 17 and correlates with high blocking temperatures there. At site 21 there is a single reversed magnetization.

At sites 14, 19, and 20, composite magnetization occurs. At site 14 the directions are southerly with moderate upward inclinations. The blocking temperatures are predominantly below 400 °C, but small magnetizations are retained at 500 °C (Fig. 9). The directions remained constant to 500 °C. This shoutherly directed magnetization is referred to as *S* magnetization. At sites 19 and 20 the thermal demagnetization shows a similar two-step curve, with some blocking temperatures below 400 °C, and others typically between 550 and 600 °C. Specimens from site 19 show this two-step curve most clearly. During thermal demagnetization dramatic changes in direction occur. The initial directions are to the south-southwest with low inclinations, at temperatures of 200 and 300 °C they have shallow negative inclinations, and then between 400 and 550 °C they move sharply to a direction with steep positive inclinations. The final direction at site

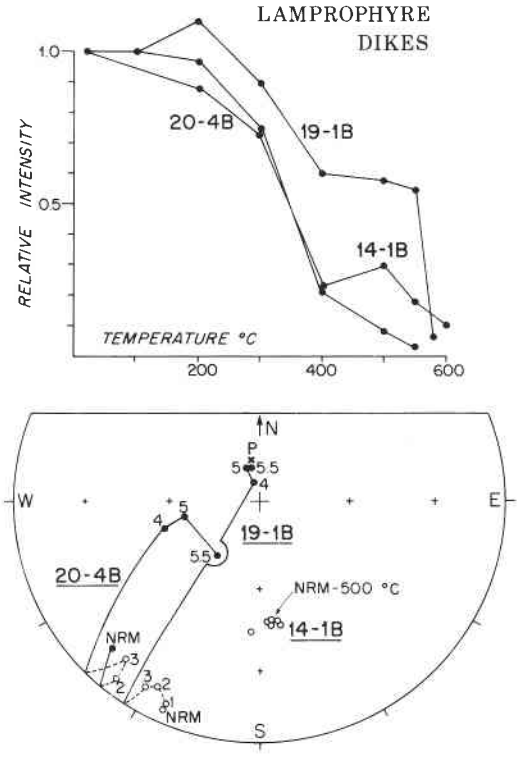


FIG. 9. Thermal demagnetization of lamprophyre dikes. Intensity changes are shown above and corresponding directions below. Dots (circles) denote downward (upward) inclination.

19 is that of a typical *N* magnetization, and the direction at site 20 approaches, but does never quite reach there (Fig. 9). The directions at sites 19 and 20 could result from a combination of *R* and *N* magnetization, but vector analysis shows that this is not the explanation (Table 2). Up to 200 °C a small vector approximately along the present field is removed. Between 300 and 400 °C a large *S* magnetization is lost. Between 400 and 550 °C very little magnetization is removed. Between 200 and 300 °C the small magnetization removed appears to be a mixture of an *S* and a present field component. Both *N* and *S* magnetizations occur in a single specimen. In other specimens from sites 19 and 20, the remanence also consists of *N* and *S*. The mean magnetization removed between 300 and 400 °C in 7 separate cores at site 19 is 197°, −38°, and at site 20 it is 212°, −23° (Table 1). These directions are similar to those at site 14 at which only an *S* magnetization is present. Thus *S* is a distinct magnetization and is not a combination

TABLE 2. Vector analysis specimen 19-1B

	Observed		Difference	
T°C	*M*	*D,I*	*M*	*D,I*
NRM	40.4	204, −05		
			1.5	024, +84
100	40.7	204, −07		
			6.7	069, +50
200	44.7	208, −13		
			9.0	195, +25
300	36.2	211, −10		
			48.1	207, −39 (= *S*)
400	24.3	333, +83		
			2.4	173, +34
500	23.3	341, +78		
			1.0	285, +68
550	22.3	345, +78		

In the first column the demagnetizing temperature is given, and in the following 3 columns the observed direction (*D, I*) and intensity (*M* in microgauss). In the last 3 columns the vector differences between each of the temperature steps are listed.

of *N* and *R*. It is characterized by blocking temperatures, which are generally below 400 °C and never exceed 500 °C. The oxide minerals visible in polished sections are similar to those of *N* magnetization sites, so that *S* is probably due to fine-grained magnetite. Thus there is no obvious difference between the minerals responsible for the *N*, *R*, and *S* magnetizations.

Matachewan Dike

Two sites were sampled in the contact aureole on Highway 11. Site 9 in amphibolite has no stable magnetization. Two thin diabase dikes occur close together in the aureole (site 10). They contain green feldspar phenocrysts characteristic of Matachewan diabase. Only a single stable magnetization is present, and the blocking temperatures lie between 550 and 600 °C. After cleaning at 550 °C the directions are close to the *N* magnetization (Fig. 4 and Table 1).

Discussion

The syenite, which comprises most of the Otto stock, has low magnetic stability and provides no information about the geomagnetic field at the time of intrusion. The lamprophyre dikes, which are late-stage products of the intrusive phase, the west-central baked zone, and Matachewan diabase within the aureole, typically have *N* or *R* magnetizations commonly with high coercivities, often exceeding 1000 Oe, and blocking temperatures between 550 and 600 °C and occasionally exceeding 600 °C. The magnetization directions of the diabase of the aureole (site 10) are very different from these of unaltered Matachewan diabase a few kilometres away (Strangway 1964; Fahrig *et al.* 1965). The dikes of the aureole appear to have been remagnetized more or less completely during the intrusion of the Otto stock, whereas those at a distance are unaffected. Hence the conditions of the contact test are satisfied, and the *N* and *R* magnetizations were probably acquired during cooling following intrusion at -2160 ± 80 m.y. One or more reversals of the field may have occurred during cooling. The mean direction and pole is listed in Table 3, generally giving unit weight to each site ($B = 18$). At site 28 both *N* and *R* can be recognized and unit weight is given to each. The pole and its error are plotted on Fig. 10. The aberrant result from site 21 (Table 1) is omitted.

The *S* magnetization is characterized by low blocking temperatures. It occurs at only three sites, and dominates only at one. The *S* magnetization could be a transitional field direction acquired during a field reversal, but it seems unlikely that such a magnetization should always be characterized by low blocking temperatures. It is suggested that this magnetization was probably acquired later than the *N* and *R* magnetizations during some subsequent mild reheating event. The pole for the *S* magnetization direction is given in Table 3 and Fig. 10, and is close to that observed in Matachewan dikes. Taken in isolation this result indicates that the stock is pre-Matachewan in age. This notion, however, is inconsistent with the fact that the magnetization of Matachewan diabase in the aureole (site 10) is close to the *N* magnetization and therefore probably reset, and with the Rb–Sr age obtained by Bell and Blenkinsop[4]. The more probable explanation is that it represents a later field direction.

The paleomagnetic poles in the interval 2200 to 1800 m.y. ago are compiled in Fig. 10. The poles form a reasonably coherent array stretching from what is nowadays the Mackenzie delta to Peru, with a possible extension farther south to include pole OS. This array is referred to as track 5 (Irving and Lapointe 1975). We suggest that the sense of apparent polar motion is from north to south. The evidence for this is that the Dubawnt Group (pole *D*), the Kahocella Formation (*K*), and the Martin Formation (*MN*) have ages between 1800 and 1900 m.y. and are younger than the Indin dikes (pole *I*), and the Otto stock. The dates of the results from the Mugford basalt, Wind River dikes, and the Otish

[4]See footnote 2.

TABLE 3. Summary of results

	B	n	Direction or pole	k	α_{95}
N+R* magnetization	18	84	330, +71	139	3
N+R* poles	18	84	69N, 133W	59	5
S magnetization	3	16	205, −30	68	15
S poles	3	16	59S, 128W	89	13

*Results calculated irrespective of sign and ommitting site 21.
B is the number of sites, n the total number of samples (two specimens measured per sample), directions are specified by declination and inclination and poles by latitude and longitude, k is Fisher's precision, and α_{95} the circle of confidence ($P = 0.05$).

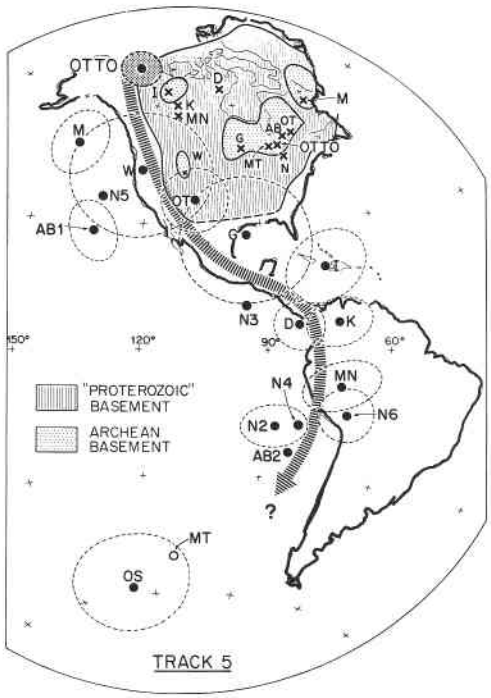


FIG. 10. Track 5. Pole positions for Laurentian Shield for the interval 2200 to 1800 m.y. with their corresponding error circles ($P = 0.05$) are given where justified. The poles and the localities they were obtained from are indexed by the same letters as follows: *AB1* and *AB2* Abitibi dikes (Larochelle 1966), 2147 ± 68 m.y. (Gates and Hurley 1973); *D* Dubawnt group (Park *et al.* 1973), 1830 ± 26 m.y.; *G* Gunflint Formation (Symons 1966), ≃2000 m.y. (Goldich 1973) *I* Indin dikes (McGlynn and Irving 1975), 2093 ± 86 (Gates and Hurley 1973); *K* Kahocello Formation (McMurray *et al.* 1973), associated volcanics 1873 ± 13 m.y.; *M* Mugford basalt (Murthy and Deutsch 1972), ≃2300 m.y. (Barton 1973); *MN* Martin Formation (Evans and Bingham 1973), K–Ar on associated dikes 1835 m.y.; Nipissing diabase (see text for age) five determinations *N2* (Symons 1970), *N3* (Symons 1971), *N4* (selected sites from Symons 1971), *N5* (Symons 1971), *N6* (Patel and Palmer 1974); *OS* Otto *S* magnetization; *OT* Otish gabbro (Fahrig and Chown 1973), ≃2000 m.y.; *OTTO* Otto stock $R + N$ magnetization; *W* Wind River dikes (Spall 1971) maximum K–Ar 2060 m.y. The pole for Matachewan diabase (*MT*) is added for comparison with *OS*.

gabbro (poles *M*, *W*, and *OT* respectively) are less well-known, but ages of about 2000 m.y. are probable.

One apparent inconsistency in this scheme is the position of the commonly accepted pole for the Nipissing diabase (poles *N2*, *N4*, and *N6*). The Nipissing diabase and the Otto stock have the same radiometric age, yet their apparent pole positions are almost 90° apart. However, the time at which the Nipissing diabase was magnetized is not definitely known. Much of the diabase, especially in the southern exposures, has been regionally metamorphosed to low green-schist facies (Card and Pattison 1973), and the descriptions of Patel and Palmer (1974) indicate that post-cooling alteration of the iron minerals could have occurred. A successful contact text has never been described. Symons (1967) has given results from the Upper Slate of the Cobalt Group which have *very* roughly the same directions as those in adjacent Nipissing diabase. This result from the Upper Slate is based on only 4 samples whose directions have a standard deviation of 40° and their mean has an error of 48° (second entry on Table 1 in Symons 1967). No conclusions can be drawn from such scattered data. Therefore it is possible that the magnetizations of the Nipissing diabase are younger that the age of emplacement. Moreover, the magnetizations observed in Nipissing diabase are very complex, and directions other than those corresponding to poles *N2*, *N4*, and *N6* occur (poles *N3* and *N5*). It is also noteworthy that the Abitibi dikes have one dominant (pole *AB1*) and one subordinate (*AB2*) magnetization, although the ages of the rocks from which they were derived are not significantly different (Gates and Hurley 1973). Poles *AB1* and *N4* are close together, as are poles *AB2*, *N2*, *N4*, and *N6*. Hence the magnetizations of the Abitibi and Nipissing diabases may be a complex mixture of as yet unresolved primary and secondarily overprinted magnetizations. In the absence of

successful contact tests and detailed thermal demagnetization of the remanent magnetization, the presently available positions of the Abitibi and Nipissing diabase poles must be regarded as uncertain, and all that can be said is that they probably lie near to Track 5. This reinterpretation of the paleomagnetic data in the interval 2200 to 1800 m.y. given in Fig. 10 is, of course, tentative. The new polar track differs very considerably from the previous interpretations, which were based on much less information (see, for instance, Irving and Park (1972), or Fahrig and Chown (1973)).

The poles of track 5 are derived from many different parts of the Laurentian Shield, including the main Archean blocks (Superior, Slave, and Nain Provinces, and the Bear-tooth uplift). The general coherence of these poles suggests that these Archean blocks did not move relative on one another by many thousands of kilometres during the Hudsonian orogeny, as has already been noted by McGlynn and Irving (1975) and Irving and Lapointe (1975). This result does not mean that relative movement did not occur, because there are still major uncertainties in the paleomagnetic data and in its time basis, for example, the unsatisfactory nature of the record from the Nipissing diabase. Nevertheless, if there had been large relative motions comparable to those among continents in the Phanerozoic, then it would be a remarkable coincidence that the poles should now form the coherent array that is observed.

Finally a note is made about the geometry of Track 5, as a preview to a discussion that cannot be satisfactorily made until more precise data become available. Relative motion in a purely longitudinal sense cannot be detected paleomagnetically; that is, if the pivot point defining the relative motion of two land masses coincides with the mean geomagnetic pole then the relative motions will have no paleomagnetic expression. This means, for example, that if relative motion of the Slave and Superior Archean blocks had occurred they would not be seen paleomagnetically if the motion had been in a longitudinal sense relative to the paleomagnetic pole. If the pivot points defining the relative motions of the Archean blocks fell on Track 5 and migrated down Track 5 following the paleomagnetic pole, then such motions would go undetected by the paleomagnetic method. The presence of a single polar track does not therefore necessarily imply that there have been no relative motions among the Archean cratons; it simply means that the hypothesis of no motions is the simplest but not the only explanation of the results.

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NOTE ADDED IN PROOF: In a very recent paper, Symons and Londry (1975) have described new results from the Nipissing diabase of the Gowganda area (47.7°N, 80.8°W). As has been found in previous studies, the magnetization is composite. There is a magnetization (*A*) with a pole (14°S, 100°W) near to previously determined Nipissing poles N2, N4, and N6, and there is a magnetization (*B*) with a pole (67°N, 111°W) near to the pole for the Otto stock. Both poles fall on Track 5 (Fig. 10). The information in Table 1 of Symons and Londry indicates that there is no significant difference between the remanent coercive forces of the *A* and *B* magnetizations. No thermal studies or studies of igneous contacts are reported. Hence from the internal evidence from the Nipissing diabase itself, it is not known whether *A* is older than *B*, or *B* older than *A*. Symons and Londry state categorically that *A* is primary and older than *B*. It is equally possible that *B* is older than *A*, and this is consistent with the new interpretation of Track 5 (Fig. 10). This interpretation has the interesting corollary that the Nipissing diabase at Gowganda and the Otto stock are essentially contemporaneous because their poles agree, and this reconciles the age evidence which also indicates that they are of the same age (see above).

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